



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences

Department Of Computer Science & Information Technology

Mid Term Examinations (Spring 2022)

Course Code with Title	MS301: Probability & Statistics	Program	BS (CS)
Instructor	Mr. Ahsan Masroor	Semester	5 th
Start date & Time	May 16, 2022 at 11:30 AM	End Time	1:00 PM
Maximum Marks	30	1 ABDUL REHMAN 215	

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt all questions that carry equal marks. Write all steps, missing steps may lead to deduction of marks.
- Please don't write anything on the question paper except your roll no and name.

Q.1.

(10)

You grew baby carrots using special soil you dig them up and measure their length (to the nearest mm).

Length mm	150-154	155-159	160-164	165-169	170-174	175-179	180-184	185-189
Frequency	5	2	6	8	9	1	6	3

Analyzing the data, estimate the following:

- Average of grouped data.
- Median of grouped data.
- Mode of grouped data.
- Compare mean, median, and mode with respect to empirical relationship and comments on your results.

SSUET

Length mm	Frequency	Midpoint	Frequency * Midpoint
150-154	5	152	760
155-159	2	157	314
160-164	6	162	972
165-169	8	167	1336
170-174	9	172	1548
175-179	1	177	177
180-184	6	182	1092
185-189	3	187	561
			$\Sigma fx = 5561$

Page 1 of 2



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences

Department Of Computer Science & Information Technology

Q.2.

(10)

The A. C. Nielsen company publishes information on the TV-viewing habits of Americans in Nielsen Report on Television. A sample of 20 people yielded the weekly viewing times, in hours, displayed in table

25	41	27	32	43
66	35	31	15	5
34	26	32	38	16
30	38	30	20	21

- Obtain and interpret the quartiles
- Determine and interpret the interquartile range.
- Find and interpret the five-number summary.
- Identify potential outliers, if any.
- Construct and interpret a boxplot

Q.3. Reasoning Questions.

(2X5=10)

- Why do we use relative frequency instead of frequency?
- Identify two main types of graphical displays that are used for qualitative data?
- Why is the standard deviation preferable to the range as a measure of variation?
- What is a frequency distribution of qualitative data and why is it useful?
- Provide a reason why the classification of data is important?

For Any Query you can add me on facebook. ☺



Ser R

Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Spring 2022)

Course Code with Title	CS-118 Web Engineering	Program	BS(IT)
Instructor	Ms. Sabeen Fatima Ms. Sana Ejaz	Semester	5 th
Start date & Time	May 19, 2022 at 11:30 AM	End Time	01:00PM
Maximum Marks	30	1 ABUL REHMAN 215	

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.

Q.No.1	(06 Marks)
--------	------------

Digital marketing is the use of social media platforms and websites to promote a product or service. Following are the popular social media to spread the word. Choose any one of the following and Give justification for your choice and enlist the unique features of each medium.

- 1) Youtube
- 2) Instagram
- 3) Facebook

Q.No.2	(08 Marks)
--------	------------

This project is an E-Commerce web application where the registered user will view and comment the product and its features. System will analyze the comments of various users and will rank product. We use a database of sentiment based keywords along with positivity or negativity weight in database and then based on these sentiment keywords mined in user comment is ranked. Comment will be analyzed by comparing the comment with the keywords stored in database. The system takes comments of various users, based on the comment, system will specify whether the product is good, bad, or worst. Once user login the system he can view the product and product features. After viewing product user can comment about the product. User can also view comment of other user's. The role of the admin is to add product to the system and to add keywords in database. User can easily find out correct product for his usage. This application also



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

works as an advertisement which makes many people aware about the product. This system is also useful for the user's who need review about a product.

Required: Draw wire frames for above project keeping in mind all the details discussed in class.

Q.No.3

(05 Marks)

Answer the following questions?

- ✓ 1. Explain basic building block of HTML and code for the basic structure of HTML.
- ✓ 2. Client runs special software (Web Browser Software), what is the purpose of this software also name the protocol which the browser uses for transmitting and receiving user requests.
- ✓ 3. Briefly explain the major components of Web?
- ✓ 4. What is the primary purpose of using "Client Server Architecture", explain it with the help of suitable diagram.
- ✓ 5. Categorized the following web application.
a. Pinterest b. CMS of SSUET c. Website of SSUET d. Microsoft365

Q.No.4

(05 Marks)

Why Requirement Engineering is considered to be important aspect when a web developer develops a website for their client? Name four steps of requirement engineering process also draw the diagram of requirement engineering process?

Q.No.5

(06 Marks)

Write the HTML code for the following login form.

Name	
Email	
Favorite Color Red:	☐
Blue:	☐
Green:	☐
Favorite Toothpaste: Crest:	☐
Close-Up:	☐
Glaxo:	☐
Click to Submit	



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science and Information Technology

Mid Term Examinations (Spring 2022)

Course Code with Title	IT-315: Enterprise Systems	Program	BS(CS)
Instructor	Ms. Anam Siddidui / Mr. Haris Ahmed	Semester	5 th
Start date & Time	May 18, 2022 at 11:30 AM	End Time	01:00 PM
Maximum Marks	30 ABDUL REHMAN 215		

Attempt All Questions:

✓ Q.1.

(06)

While ERP provides a number of obvious benefits, this is not without challenges. Discuss three major categories of challenges faced by an organization while implementing an ERP system.

✓ Q.2.

(06)

Designing proper roles and authorizations in ERP system ensures that every user using the system have right authorizations nothing more, nothing less. Explain activity-based authorization with an example.

✓ Q.3.

(05)

“ SAN FRANCISCO, May 10, 2022 - PR Newswire: The global ERP software market size is anticipated to reach USD 123.41 billion by 2030, registering a CAGR of 10.7% during the forecast period, according to a new report by Grand View Research, Inc.”
 Discuss the reasons of explosive growth of ERP market and ERP vendors.

✓ Q.4.

(05)

ERP is about integrated information system that needs an integrated database. All departmental databases are replaced by a single integrated relational database for the entire company. Describe the advantages of a corporate wide integrated database.

✓ Q.5.

(08)

The objective of data cleansing is to make data accurate, consistent, complete and useful. While data can be always cleaned manually, it sometime becomes practically impossible to clean large database manually. Discuss different algorithms that can be used for auto cleaning of data.



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Spring 2022)

Course Code with Title	HS-401: Technical Report Writing & Presentation Skills	Program	BS(CS)
Instructor	Tazeen	Semester	5 th
Start date & Time	May 17, 2022 at 11:30 AM	End Time	01:00PM
Maximum Marks	30		

See E.

ABDUL REHMAN 215

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.
- Mention your Section on top of the answer script.

Q.1.

Paraphrase the following paragraph by using the effective tactics of paraphrasing.

(10)

Big data can be defined as data sets that are too gigantic and intricate to be analyzed with the conventional data analyzing software, tools, and techniques. The three main characteristic features of big data are volume, velocity, and variety, where volume represents the huge amount and mass of data generated, velocity represents the rate at which these data are being reproduced, and variety represents heterogeneity present in the data sets. With the advent of microarray, RNA-seq, and high-throughput sequencing (HTS) technologies, a plethora of biomedical data is being engendered every day, due to which contemporary drug discovery has made a transition into the big data era. In drug discovery, the first and foremost step is the identification of appropriate targets (e.g., genes, proteins) involved in disease pathophysiology, followed by finding suitable drugs or drug-like molecules which can meddle with these targets, and now we have access to a constellation of biomedical data repositories which can help us in this regard.



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Q.2.

(10)
Make a list of all the elements of effective report writing. Explain them in detail along with its importance in a technical report in your own words.

Q.3.

(10)
(a) Write down the difference between in-text citation and parenthetical citation along with the appropriate examples.

(b) Why do you have to reference? Identify the given reference and write down the following elements of the reference in the correct order with example, according to the Harvard Style.
(Word Limit: 100-150)

- Anderson. B.
- *Reflections on the origins and spread of nationalism.*
- *Imagined communities.*
- Verso.
- (2006)

Plagiarism

①. correctness.

②. accuracy.

③. comprehensiveness.

④. Relevance → ?

⑤. precise



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Spring 2022)

Course Code with Title	CS-310: Compiler Construction	Program	BS(CS)
Instructor	Ms. Ayesha Urooj Mr. Rajkumar Chawla	Semester	5 th
Start date & Time	May 19, 2022 at 11:30 AM	End Time	May 19, 2022 01:00PM
Maximum Marks	30	1 ABDOUL REHMAN 215	

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt all questions.
- Use of calculator is not allowed.
- Write your answers in ink. For drawings pencils may be used.

Q.1(a).

(05)

Illustrate the number of steps when skeleton source program enters and turns into absolute machine code. Support your answer with the help of block diagram.

Q.1(b).

(05)

Show the output of each phase when the given input string is passes through the compiler.

Input String: $a+b*c$

Q.2.

(10)

Write all basic requirements that perform in Syntax Analysis phase for the following code.

Begin

int a=1, b=5, c;

c = a * b + c;

Print c;

End.



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Q.3

(04)

Prove the ambiguity in the following grammar?

$E \rightarrow E * E$

$E \rightarrow E + E$

$E \rightarrow id$

Q.4

(06)

Write the grammar, show the derivation and draw the parse tree for the following:

(i) $-(id * (id + id) - id)$

(ii) $(id * id) - (id / (id + id))$

$- (E * (E + E) - E)$



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science and Information Technology

Mid Term Examinations (Spring 2022)

See R

Course Code with Title	CS-317T: Computer Networks	Program	Bachelor of Computer Science (BSCS)
Instructor	Razia Nisar Noorani/ Tahmina Khan	Semester	5 th
Start date & Time	May 21, 2022 at 11:30 AM	End Time	1:00PM
Maximum Marks	30	1 ABDUL REHMAN 215	

IMPORTANT INSTRUCTIONS:

Read the following Instruction carefully:

- Attempt all 5 Questions. Each question carry equal Marks.

Q.1.

(3+3)

- (a) Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable? Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are 1.5Mbps and 1.2Mbps, respectively. Assuming that the switch uses store-and-forward packet switching, what is the total end-to-end delay to send a packet of length 100Mb? (Ignore queuing, propagation delay, and processing delay).
- (b) UDP uses 1's complement for their checksums. Suppose you have the following three 8-bit bytes: 01010011, 01100110, and 01110100. What is the checksum byte generated from sender side. Show all work. Why is that UDP takes the 1's complement of the sum; that is, why not just use the sum? With the 1's complement scheme, how does the receiver detects errors?

Q.2.

(3+3)

- (a) Suppose Ahmed, with a Web-based e-mail account (such as Hotmail or Gmail), sends a message to Bilal, who accesses his mail from his mail server using POP3. Discuss how the message gets from Ahmed's host to Bilal's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts. (Also draw the whole conversion).



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science and Information Technology

- (b) Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?

Q.3.

(06)

Justify the two statements below:

- (1) With non-persistent connections between browser and origin server, it is possible for a single TCP segment to carry two distinct HTTP request messages. (YES/NO with one line justification).
- (2) If a user requests a Web page that consists of some text and three images. For this page, will the client send one request message and receive four response messages (YES/NO with one line justification).

Q.4.

(06)

Consider a planet where everyone belongs to a family of six, every family lives in its own house, each house has a unique address, and each person in a given house has a unique name. Suppose this planet has a mail service that delivers letters from source house to destination house. The mail service requires that (1) the letter be in an envelope, and that (2) the address of the destination house (and nothing more) be clearly written on the envelope. Suppose each family has a delegate family member who collects and distributes letters for the other family members. The letters do not necessarily provide any indication of the recipients of the letters. Compare the scenario with the transport layer. Clearly define the roles.

- (1) Describe a protocol that the delegates can use to deliver letters from a sending family member to a receiving family member.
- (2) Compare the scenario with the transport layer. Clearly define the roles.

Q.5.

(3+3)

- (a) Multiplexing/Demultiplexing happens at all layers. Justify the statement with proper proof? What is an application-layer message? A transport-layer segment? A network layer datagram? A link-layer frame?
- (b) What is the difference between connectionless demultiplexing and connection oriented demultiplexing. Draw the diagram for apache server sending and receiving messages to client1 and client2 with all information needed for connection oriented demultiplexing.

(END OF PAPER)

For Any type of Query you can ask on facebook
or whatsapp me.



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Faculty of Computing and Applied SciencesDepartment: Department of Computer Science and Information TechnologySemester 5th Degree Program BS(CS) Batch 2020SCourse Code: IT-315 Course Title: Enterprise Systems Credit Hours: 3+0

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	<u>215</u>	Name	<u>[Signature]</u>	'A' Copy Serial No.	<u>110433</u>
----------	------------	------	--------------------	---------------------	---------------

Q.No.1

Marks 15

Siebel was the undisputed market leader in CRM upto Year 2005 and one of the first vendors to make CRM software popular in the enterprise market. Oracle had acquired Siebel Systems in January, 2006. Discuss in detail what are the key features of Oracle Siebel CRM?

Q.No.2

Marks 15

Master data in an ERP system are the data records that are stored in the database centrally for a long period of time and are used by multiple modules of ERP system. Explain different master data for ERP procurement module. What are material master and vendor master?

Q.No.3

Marks 10

Data Mart is a subset of a data warehouse that supports the requirements of particular department or business function. Discuss what are the characteristics and advantages of a data mart? How a data mart is different from data warehouse?

① OP m

② OL m

③ Acc m

④ Sales

⑤ marketing

Auto

Email

~~For~~ ~~not~~ ~~out~~



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Q.No.4

Marks 05

- ✓ B2B Portals helps to build better relationships between the company and its suppliers, customers and partners (via extranets) and this improved relationship can lead to increased trading partner loyalty. Explain the different types of B2B portals?

Q.No.5

Marks 05

An operational data store (ODS) consolidates data from multiple source systems and provides a near real time integrated view of current data. Discuss the similarities and differences between a data warehouse and an ODS?



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Exam Date: 27-06-2022

Faculty: Computing and Applied Sciences Department: CS & ITSemester: 5th Degree Program: BS (CS) Batch: 2020SCourse Code: MS301 Course Title: Probability & Statistics Credit Hours: 3+0

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	<u>215</u>	Name	<u>A. Raza</u>	'A' Copy Serial No.	<u>100109</u>
----------	------------	------	----------------	---------------------	---------------

Q.No.1

10 Marks

An important concern for any pharmaceutical company producing drugs is to determine how a particular drug will affect one's perception or general awareness. Suppose one such company wants to establish a relationship between the PERCENTAGE of a drug in the blood-stream and the LENGTH OF TIME it takes to respond to a stimulus. Suppose the company administers this drug on 5 subjects and obtains the following information:

Subject	Percentage of drug	Reaction Time (milli-seconds)
	X	Y
A	1	1
B	2	1
C	3	2
D	4	2
E	5	4

- Determine the regression equation for the data.
- Graph the regression equation and the data points.
- Determine the sums of squares in regression, SSR
- Determine the Total sum of squares, SST
- Determine the Coefficient of Determination

$$SSR = y - \hat{y}$$

Q.No.2

10 Marks

Suppose that the population of interest consists of the five starting players on men's basketball team whom, for convenience, we will call A,B,C,D and E. Further suppose that the variable of interest is height, in inches table below list the player and their heights.

Player	A	B	C	D	E
Height	76	78	79	81	86



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Exam Date: 27-06-2022

- I. Verify that mean of all sample means is equal to population mean
- II. Find out standard deviation of population and sampling
- III. Determine standard error.

Q.No.3

10 Marks

- I. A class contains 16 boys 10 girls of which half of the boys and half of the girls are fat. Find the probability that a student chosen at random is a boy or fat student.
- II. A box contain 8 ticket bearing the number 1,2,3,4,5,6,8,10. One ticket is drawn and kept aside. Then a second ticket is drawn. What is the probability that both the tickets shown even numbers

Q.No.4

10 Marks

A random sample of 100 workers with children in day care show a mean day-care cost of Rs.2650 and a standard deviation of Rs.500. Verify the department's claim that the mean exceeds Rs.2500 at the 0.05 level with this information.

Test	α	0.01	0.02	0.05	0.10
Two tail	$Z_{\alpha/2}$	2.576	2.33	1.96	1.645
One tail	Z_{α}	2.33	2.05	1.645	1.28

Q.No.5

10 Marks

- I. The probability that a patient survives from a delicate heart operation is 0.85. What is the probability that exactly 3 of the next 4 patients having this operation survive.
- II. A secretary makes 2 errors per page on the average. What is the probability that she makes a) no error b) 5 error on the next page she types.





Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Computing & Applied Sciences Department: Computer Science & Information Technology
Semester: 5th Degree Program: BSCS Batch: 2020S
Course Code: HS-401 Course Title: Technical Report Writing & Presentation Skills Credit Hours: 2+0
Timings: 3 Hours Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	<u>215</u>	Name	<u>A. Ru</u>	'A' Copy Serial No.	<u>105299</u>
----------	------------	------	--------------	---------------------	---------------

Q.No.1

Marks 10

✓ During an interview, as an interviewee, we make a number of blunders in front of the interviewer. If you go for an interview for a 9 to 5 job, for a front desk position, how would you overcome your dressing issues and perform well during an interview?

Q.No.2

Marks 10

How would you practice assertiveness as a professional when unnecessary or excessive favors are asked of you when you manage a team of 20 employees with multiple tasks to finish and deadline approaching near? Elaborate your response with the strategies that are essential to practice in such a scenario.

Q.No.3

Marks 10

✓ Aisha has a strong background in social science research and evaluation. She manages a staff that works on different projects within the agency. She is known as a problem solver and is extremely supportive of her staff. The managers are sensing that the staff is becoming overworked as everyone takes on increased responsibilities due to high staff turnover. They are having difficulty in appropriately allocating work to the staff. Aisha said that she had not received sufficient information from the finance department to complete the budgets.



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Keep in view the given scenario, if you were the manager and had to deliver bad news to your team, how would you have performed well with such a situation?

Q.No.4

Marks 10

Microsoft made a shocking announcement that it was buying \$7.2 billion from Finnish mobile handset manufacturer Nokia, a merger aimed at building the mobile and Smartphone products of Microsoft. After the ink dried on the deal, the merger encountered even more difficulty, namely, the difficulties of integrating workers from various cultures.

According to the aforementioned case, how would Microsoft propose the significance of ideas during a table negotiation? Validate the situation through authentic steps when you negotiate with others in the organization to settle an agreement.

Q.No.5

Marks 10

During a presentation, it is necessary to maintain your gestures, postures and body language. Do you agree or disagree? Answer in detail and also state reasons to prove your point.

Pre-neg
nego.
Set-on
conclusion



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Computing & Applied Science Department: CS & IT

Semester: 5th Degree Program: BS (CS) Batch: 2020S

Course Code: CS-311 Course Title: Web Engineering Credit Hours: 2+0

Time: 2:00 to 5:00 P.M. Date: 04-07-2022

Timings: 3 Hours Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	<u>AP 215</u>	Name	<u>As</u>	'A' Copy Serial No.	
----------	---------------	------	-----------	---------------------	--

Q.No.1

10 Marks

Suppose you are working in an organization and carry the responsibilities of "Web Developer". Your aim is to design the website for company's client who is based on certain requirements including tools and functional and non-functional requirements. The title of the website is "Online Exams", which is a dynamic website and contain the information related to conduction of examination including question papers, uploading and downloading of papers. Being the web developer give the answers of the following questions.

- By galk down*
- What do you think about the requirements you will gather from your client?
 - List down the steps through which a Web Engineer has to go through to develop a website.
 - Discuss aesthetic design suitable for this scenario?
 - When you first meet the client what questions you will ask?
 - Elaborate Requirement Elicitation Process for this scenario?
 - Comment about its functional and nonfunctional requirements?
 - Which website features you will develop at initial prototype?

Q.No.2

10 Marks

- ☒ a. Create a webpage having header, contents, navigation and footer sections. Header should contain the logo and/or name of your webpage and footer should have the copyrights statement. Display the contents and navigation of your own choice. Use bootstrap to design the above layout.

OR

Use HTML and CSS to design the above required Web page.

- b. Design XML file to hold data for the following orders.
- Order #1.

Quantity	Description	Unit Price	Order Price
2	Young's chicken spread	350	Rs.700
1	American Garden 's Tomato Puree	200	Rs.200
2	Lipton Yellow Label Tea	240	Rs.480

Order #2.

Quantity	Description	Unit Price	Order Price
2	Young's chicken spread 1/2 kg	240	Rs.480
1	Hilal's chocolate cupcake box	140	Rs.140



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Q.No.3 5 Marks

- What is the keyword for detecting that the user is hovering over an element?
- Discuss the difference between http and https with example.
- Is there any advantages of cookies? What is the max amount of data can be stored by cookies?
- Discuss the difference and advantages of JavaScript and jquery with suitable example

Q.No.4 10 Marks

- The list items are the navigation for a site. They should be laid out as a row, with an equal amount of space between each item. (Flex Box)



- Use blur(), click() and keypress() events in your program.

Q.No.5 15 Marks

ABC company wants to get feedback about their services. For this you have been asked to design a feedback form as shown in diagram below.

- Write html code for the following form in a page named **subscription.html**
- Submit form values to a page named **subscribed.php** save results in the table subscription of the database name **DB** and make a file named **connect.php** for making database connection.

Newsletter Subscription:

Please fill in the fields below to subscribe to the newsletter.

Fields marked with asterisk () are required*

*Name:

*Email:

Gender: ☒ Male ☐ Female

City:

Province:

Country:

☐ I have understood and agreed to the Terms & Conditions.

SUBMIT

\$Server name
\$url name
\$password



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Computing and Applied Sciences Department CS&IT

Semester 5th Degree Program BS(CS) Batch 2020S

Course Code: CS-317T Course Title: Computer Networks Credit Hours: (3+0)

Date: 08-07-2022
Timings: 3 Hours

Time: 2.30 to 5.30pm
Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	Name	'A' Copy Serial No.

Q.No.1

5+5

- a) Suppose Host A sends two TCP segments back to back to Host B over a TCP connection. The first segment has sequence number 90; the second has sequence number 110.
- How much data is in the first segment? 19
 - Suppose that the first segment is lost but the second segment arrives at B. In the acknowledgment that Host B sends to Host A, what will be the acknowledgment number?
- b) For reliable data transfer rdt 3.0 Draw and Discuss in detail the packet loss, Acknowledgement loss and Delayed/Premature timeout scenarios.

Q.No.2

5+5

- (a) How Netflix works describe in detail with proper diagram of content distribution servers.
- (b) In your opinion what are the advantages of Peer to Peer (P2P) Network? What is a P2P network used for? Explain the working of Bit Torrent.

Q.No.3

5+5

- (a) Host A wants to send 10 frames to Host B. The Hosts agreed to go with Selective repeat having window size 4. How many numbers of frames are transmitted by Host A if every 6th frame that is transmitted by Host A is either lost or corrupted? Show the step by step progress by drawing and writing the details.

Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

- (b) Station A need to send a message consisting of 9 packets to station B using a sliding window (window size 3) and go back n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACK from B ever lost), then what is the number of packets that A will transmit for sending the message to B? Show every step with properly drawing the sliding window received and lost packets.

Q.No.4

5+5

- (a) Describe and differentiate the two most important network-layer functions in a datagram network? What is the 32-bit binary equivalent of the IP address 223.1.3.27?
- (b) Suppose there are three routers between a source host and a destination host. Ignoring fragmentation, an IP datagram sent from the source host to the destination host will travel over how many interfaces? How many forwarding tables will be indexed to move the datagram from the source to the destination? DRAW the above scenario and describe in detail.

Q.No.5

6+4

- (a) Find Shortest path from node a to all the other nodes using Dijkstra's Shortest Path Algorithm for image 1 and Find shortest path from node S to all the other nodes using Bellman ford Algorithm in image (2).

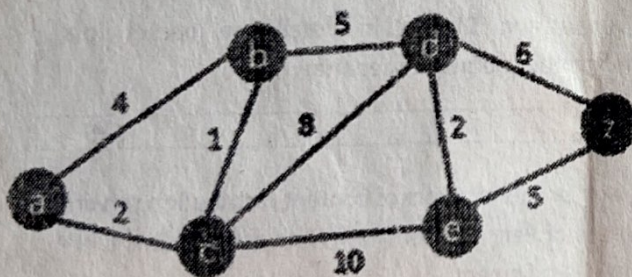


Image (1)

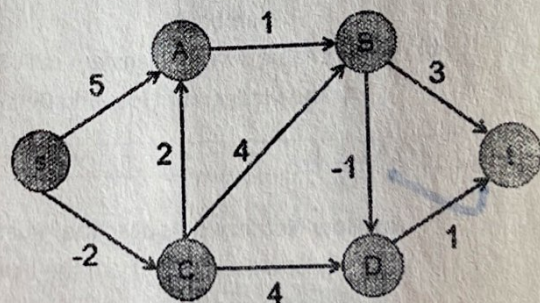


Image (2)

- (b) Describe in detail the following classes of MAC Protocol:
- Channel partitioning
 - Random Access
 - Taking turns



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Faculty of Computing & Applied Sciences Department: Computer Science & Information Technology

Semester: 5th Degree Program: BS-CS Batch: 2020S

Course Code: CS-310 Course Title: Compiler Construction Credit Hours: 3+0
Timings: 3 Hours Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	<u>215</u>	Name	<u>A</u>	'A' Copy Serial No.	
----------	------------	------	----------	---------------------	--

Q.No.1(a) **07 Marks**
Explain the responsibilities of a parser and explain its role in compiler model with the help of block diagram?

Q.No.1(b) **03 Marks**
Write translation schema for the following

- a> $S \rightarrow \text{if } E \text{ then } S1$
- b> $E \rightarrow E1 \text{ mod } E2$
- c> $E \rightarrow E1 [E2]$

Q.No.2 **10 Marks**
Construct a Context Free Grammar which can generate a program written in your own language. Program should start with the keyword which shows the beginning of code and ends with the keyword of your own choice. It generates the complete mark sheet of the student.
You have to enter your complete roll# so that it will display your mark sheet of current semester, display the message of Pass/Fail and if your percentage is less than 50% you are considered as FAIL.

User define functions, multiple declaration, Conditional Statements, loops must be covered in order to achieve the above mentioned scenarios.



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Q.No.3**10 Marks**

By considering the concept of Non-Recursive Predictive Parser, check whether the given grammar is LL(1) or not? If not then explain the reason.

$E \rightarrow bETS \mid bETSaS \mid x$
 $a \rightarrow y$

Q.No.4**10 Marks**

Implement Bottom Up Parsing Technique SLR for the given grammar.

$E \rightarrow E - E$
 $E \rightarrow E / E$
 $E \rightarrow \text{num}$
 $E \rightarrow (E)$

Q.No.5(a)**05 Marks**

Draw the dependency graph for the given grammar by taking string of your own choice

$D \rightarrow TL$
 $T \rightarrow \text{int} \mid \text{real}$
 $L \rightarrow L1, id \mid id$

Q.No.5(b)**05 Marks**

Construct the syntax tree for the expression $a - 4 + d$ and write the semantic Rules for the given grammar

$E \rightarrow E + T \mid E - T \mid T \mid id \mid \text{num}$